FOODSHIELD AI

# The structural-risk layer beneath global food markets

One auditable risk score per country · 193 countries + 50 US states · 33 public data pipelines, every input traceable

**Prepared for RaboResearch Food & Agribusiness**
Fedde Vreeken · Erasmus University Rotterdam · June 2026

WHO I AM

# A student who got curious — not a food-security expert

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**I built FoodShield AI solo, with AI-assisted coding and review.**

I'm not a food-security specialist. I started it because of one question: if something breaks in one country tomorrow — a war, a drought, a port closure — who actually gets hurt, and how badly?

The data to answer that exists — across FAO, the World Bank, WFP, USDA, ACLED and more — but it's scattered across agencies, formats and APIs. Nobody had wired it together into one comparable view. So I did.

THE GAP

# Deep research exists. The cross-country layer doesn't.

## TODAY

- 150+ RaboResearch analysts produce deep, siloed work
- A wheat outlook here, an FX note there, a fertiliser piece
- Excellent depth — but no single comparable view across countries

## FOODSHIELD FILLS

- One 0–100 risk score per country, same method everywhere
- Fuses trade + climate + conflict + governance + prices
- Every number traceable to its public source

WHAT IT IS

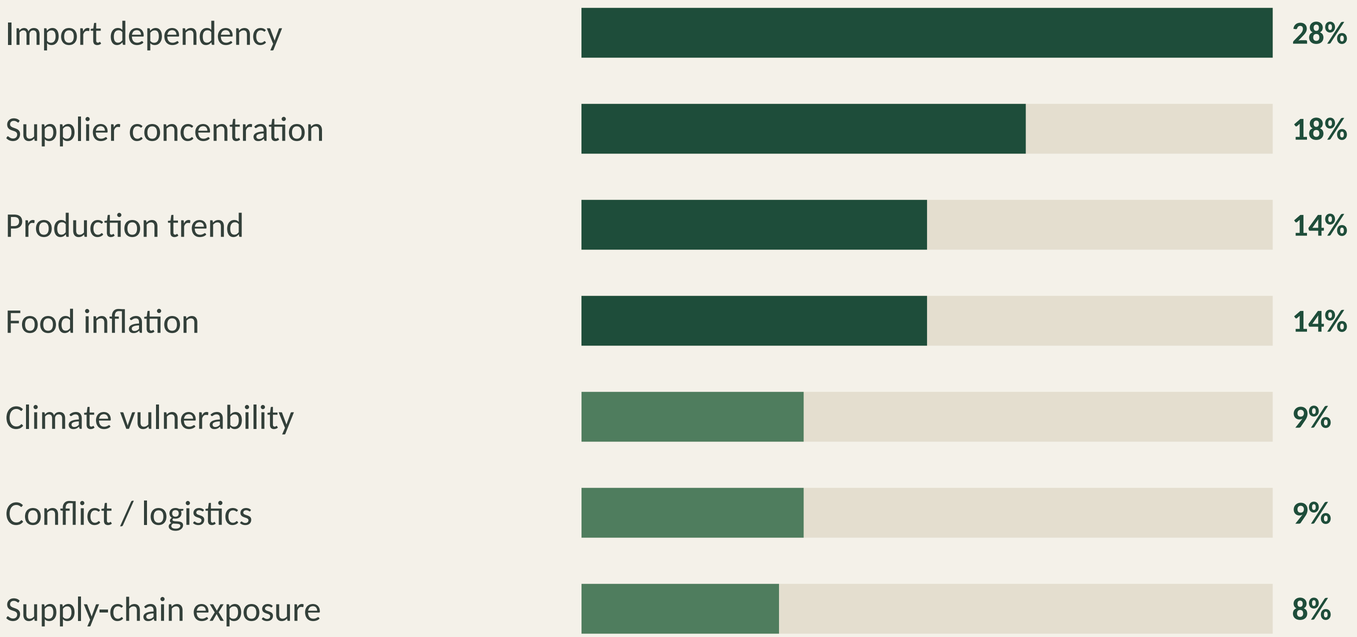
# One score. Seven components. Fully sourced.

| 243 | 33 | 7 | 0–100 |
| --- | --- | --- | --- |
| **places scored** | **data pipelines** | **weighted components** | **FDRS score** |
| 193 countries + 50 US states | refreshed every 6 hours | 28·18·14·14·9·9·8 | structural + live nowcast |

*FDRS = Import Dependency · Supplier Concentration · Production Trend · Food Inflation · Climate · Conflict/Logistics · Supply-Chain Exposure*

# How the FDRS score is built

Two layers per country: a structural baseline, plus a live nowcast adjustment from current signals.




**STRUCTURAL**

Slow-moving baseline from sourced + heritage inputs.

**+ NOWCAST**

Live −10…+35 overlay from IPC, prices, FX, weather, conflict.

**= HEADLINE 0–100**

Every weight documented; every input traceable.

COMPONENT · TRADE FLOW ATLAS

# Where the world's food goes — and the chokepoints

Per-commodity import/export flows, supplier concentration, and chokepoint risk — with each route labelled observed or modeled.

EXAMPLE · CHOKEPOINT

**43%** of Egypt's wheat is from Russia

HHI supplier concentration 72 — no diversified fallback in the top 3 suppliers. A single trade-route disruption would expose Egypt's wheat supply within weeks.

- **Supplier concentration**

  HHI index + ranked share donut, per commodity

- **Chokepoint flags**

  Single-source dependencies surfaced first

- **Observed vs modeled**

  Green = real UN Comtrade bilateral; amber = estimated

- **Net trade position**

  Importer / exporter balance per country

# Ten commodities, ~85% of traded food calories

Live benchmark prices, concentration scores, and the chokepoint behind each — in the language of a commodity desk.

### Wheat

**$246**

↓0.4% MoM

Conc 40

### Soybeans

**$463**

↓2.0% MoM

Conc 88 · alert

### Palm oil

**$1,148**

↑3.6% MoM

Conc 85 · alert

### Fertilizer

**$857**

↑18.1% MoM

Top mover

*Prices: World Bank Pink Sheet (monthly, $/MT) · Concentration: UN Comtrade + USDA PSD · every card badged observed or modeled.*

COMPONENT · SCENARIO STRESS TEST

# What if a major supplier stops exporting?

Layer a real shock on top of the structural model — a wheat embargo, a drought, a conflict — and see which countries break first.

### 1 · Pick a shock

Wheat embargo · drought · port closure · conflict flare-up

### 2 · Choose severity

Each shock maps to the FDRS component it stresses

### 3 · See who breaks

Countries re-ranked by their post-shock score, most exposed first

*A structural what-if — not a price forecast. The combining rule (√ of summed shocks on the same channel) avoids double-counting.*

UNDER THE HOOD

# 33 pipelines → one score, every 6 hours

1

### 33 public feeds

FAO · World Bank · WFP · USDA · ACLED · INFORM · CCKP · Comtrade...

→

2

### Normalise

Each feed → stable JSON, retried, provenance-tagged

→

3

### Compose

Structural FDRS + nowcast overlay + climate/governance blend

→

4

### Serve

Static dashboard, no account, refreshed automatically

*Static frontend + Python data layer, automated via GitHub Actions. No server, no database, no account.*

THE DIFFERENTIATOR

# Every number says how much to trust it.

A research desk trusts honest scope over a polished black box. FoodShield labels every value — and shows what it doesn't know.

| | |
|---|---|
| **SOURCED** | Verified against a live public dataset — source & date shown |
| **MODELED** | Estimated from structural patterns — clearly flagged, never as fact |
| **LEGACY** | Heritage baseline, not yet re-verified — badged, not hidden |

***Live source health is shown as 24 / 33 — not a fake 33 / 33.***

SOURCED & CURRENT

# The structured data a desk would otherwise build by hand

**Food inflation**

Eurostat HICP · monthly · 31 countries sourced

**Governance**

World Bank WGI · all 6 dimensions · −2.5…+2.5

**Observed climate**

WB CCKP · recent-decade warming vs 1991–2000

**Caloric trade shares**

FAOSTAT Food Balance Sheets

**Commodity prices**

World Bank Pink Sheet · $/MT · monthly

**Humanitarian risk**

EU JRC INFORM 2026 · 191 countries

HONEST LIMITATIONS

# What it does not claim

Said before you ask — owning these is the point of a risk tool.

### Not a forecast

The 2030 outlook is an illustrative scenario, not a back-tested prediction — and is labelled as such.

### Structural baseline is partly heritage

Some country fields are hand-curated, not yet re-sourced. All clearly badged.

### Some crisis feeds are upstream-down

WFP HungerMap & IPC return server errors today. The dashboard degrades honestly — affected countries flagged low-confidence, never faked.

HOW IT WAS BUILT

# AI as pair programmer — the judgement stayed mine

The dashboard asks you to trust a synthesis of public sources. The build needed the same discipline: sources checked, claims narrowed, modelled values labelled.

### AI HELPED WITH

- Writing & refactoring the 33 data pipelines
- Catching bugs as a critic — e.g. an empty-string supplier match that inflated Egypt's wheat exposure; centroid fixes for misplaced countries
- Implementing & stress-testing the scoring code

### I DECIDED

- The seven components and their weights
- The split between structural risk and live nowcast
- Which sources to trust, and the limits on modelled claims
- **AI helped implement and critique the system — it did not decide what the score should mean.**

WHAT I LEARNT

# Three things the build taught me

1

### The synthesis is the hard part

Each public source is excellent on its own. The value — and the difficulty — is wiring 33 of them into one comparable, traceable view. The data is free; the integration is the product.

2

### Honesty beats polish

It's tempting to make every number look authoritative. A risk tool is more trustworthy when it shows what it doesn't know — provenance badges, 24/33 source health, low-confidence flags when crisis feeds are down.

3

### Concentration is the real story

The world's food trade is far more concentrated than I expected — a handful of suppliers feed most of the countries already in crisis. That single insight reshaped how I weighted the score.

THE ASK

# Could a desk like yours use a structural-risk layer beneath its market work?

- Which single public source would sharpen one component enough to justify it?
- Where would this lose an analyst's trust?
- **Complementary to your price & forecast work — not a competitor to it.**